Jason Yang

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA

Candidate for B.S. in Electrical Engineering with Computing May 2029

- Relevant Coursework: *Intro to C & Assembly, Computation Structures, Linear Algebra & Optimization, Signal Processing, Math for CS (Discrete), Intro to Algorithms*

Projects

Low-Power Desk Display June 2026 - Present

- Designed and built a custom PCB integrating an ESP32-S3 and power regulation, interfacing with a Raspberry Pi and display to create a compact holographic-style desk display featuring weather, music playback, calendar tasks, and Bluetooth headphone status
- Developed embedded firmware, graphics rendering, and a custom communication protocol over SPI with acknowledgement signaling for reliable cross-device communication

String Art Robot October 2025 - December 2025

- Collaborated on a robot that recreated digital images by routing string across a board of 200 nails
- Developed C++ image processing algorithms and tooling to generate optimized stringing paths and control an Arduino-based stringing robot, successfully drawing multiple pieces, including the Mona Lisa and the MIT Dome

Experience

MIT Space Propulsion Lab Cambridge, MA

Undergraduate Researcher May 2026 - Present

- Developing mission control and flight software for a 3U-XL CubeSat designed to validate staged electrospray propulsion technology for deep space exploration
- Conducting end-to-end satellite integration and system-level verification of hardware, software, and subsystem interfaces, demonstrating successful L-band telemetry and command transfer

MIT Rocket Team Cambridge, MA

Controls Lead August 2025 - Present

- Implementing closed-loop throttle control for a bipropellant liquid engine, achieving mass-flow settling times under 5 seconds in simulation while maintaining the target mixture ratio
- Writing embedded drivers and flight firmware for a propeller-powered VTVL hopper, integrating a 50 Hz state estimation loop with a modular flight-control interface
- Developing a hardware-in-the-loop framework integrating flight hardware with simulated vehicle dynamics to validate sensor interfaces, state estimation, and control algorithms for a propulsive lander

MIT Media Lab - Responsive Environments Cambridge, MA

Undergraduate Researcher August 2025 - Present

- Reverse-engineered firmware for a smart BLE wearable ring to enable real-time sensor data streaming and integrated TensorFlow for gesture recognition
- Prototyping a dense modular capacitive/resistive sensor array for use on fabrics

The RoboLancers Philadelphia, PA

Programming Lead, Project Manager June 2023 - June 2025

- Led a team of 8-10 programmers, developing award-winning robots emphasizing maintainability and broad team participation
- Developed a programming curriculum, building technical competency across team of 50+ members
- Co-led a project management effort for 100+ members to streamline core workflows, reducing initial robot build time by over 10%

Skills

C, C++, Python, Java, JavaScript/TypeScript, STM32, Embedded Systems, Git, KiCad